

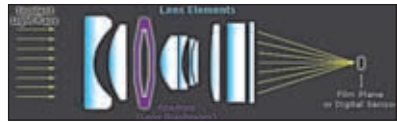
move on to other features. Vibration Reduction feature might be an exception here in terms of importance. You'll find the focal range (or the excel focal length in case of prime lenses) along with aperture range length mentioned on the box.

So a 18-55mm, f3.5/5.6 lens means that at 18 mm of focal length, it can open up to a maximum f-number 3.5 and when you zoom in to the 55mm point the maximum opening up of the lens would be reduced to f/5.6.

Lens elements and image quality

Lenses are often complex optical assemblies composed of many “lens elements.” Each of these elements have a specific role to play in focusing the path of light rays to the sensor in order to recreate the image as accurately as possible with the least amount of distortion. The distortion can be in terms of chromatic aberrations or even in terms of dimensions. In lenses with Vibration Reduction or Optical Image stabilisation, one or many lens elements would be involved in that process.

Optical aberrations occur when points in the image don't translate back onto single points after passing through the lens.



All working together to give you a great result

This tends to lower the image quality by blurring, reducing contrast or mis-aligning colours. Lenses may also suffer from uneven, radially decreasing image brightness (vignetting, of course, when not used as a feature) or distortion.

Any of the above problems is present to some degree with any lens. It's always a good idea to read online reviews about the lens that you plan to purchase, check out sample shots and look for these optical characteristics.

Kit lens

These days the lens that's offered as a package with the dSLR is generally called the “kit lens”. It's generally an entry-level quality zoom lens. This is usually a fairly general purpose lens designed for everyday shooting. On most low-end dSLRs you'll find this to be an 18-55mm lens, though on higher-end bundles you might find a 18-105mm bundled along. Though conventional advice goes that kit lenses are of lower quality and you should always prefer to buy the dSLR as a body only and to buy your own selection of lenses, we'd say that these days some of the kit lenses bundled along form brilliant company; for example, the Nikon 18-55mm VR II lens available with most of its cameras is a sure gem in terms of features and performance.